Model to Market - Evidence Pack

An AI-native, self-adapting quantitative trading system

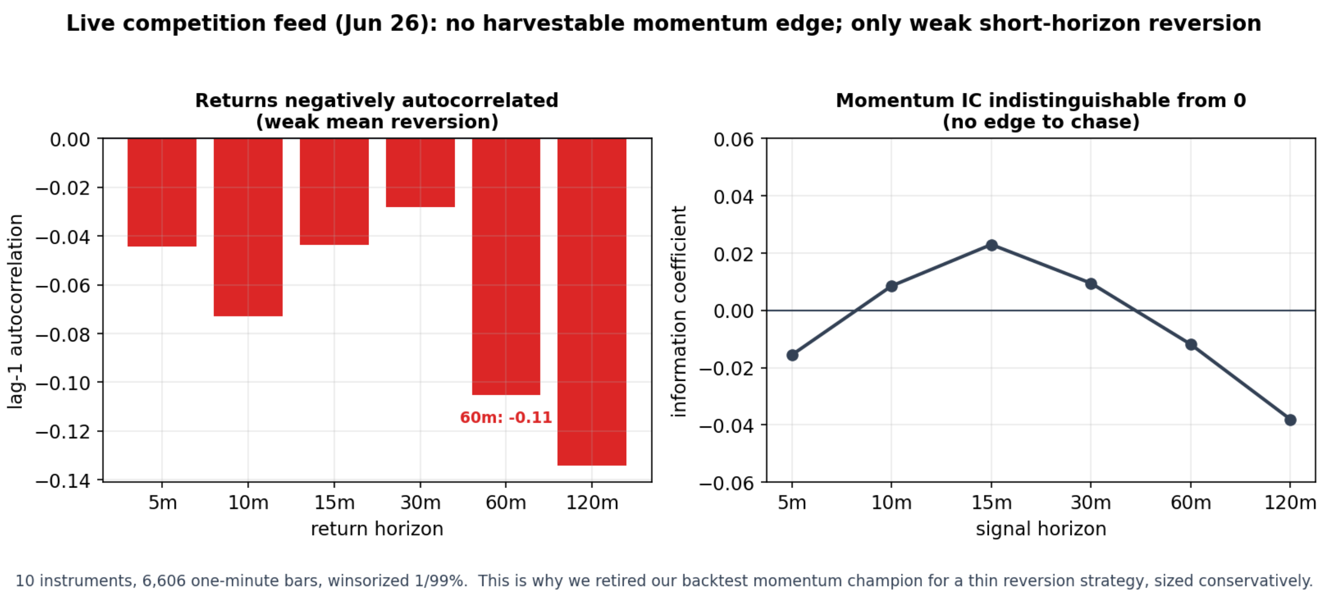
Alex Lahdiri | alex@ladimex.com | Account 10009 | $1,000,000 simulated | scored 70 / 15 / 10 / 5
Repository: github.com/Alex-Lahdiri/Quanthack-Model-to-Market

I built a self-adapting, governed multi-agent trading desk. Each cycle, NVIDIA Nemotron classifies the live regime and Anthropic Claude proposes the next strategy, leverage and cadence - every proposal a Pydantic-validated object - and a deterministic Governor then clamps it to competition-safe limits before it can act. AI proposes, the governor disposes, and Logfire traces every decision end to end.

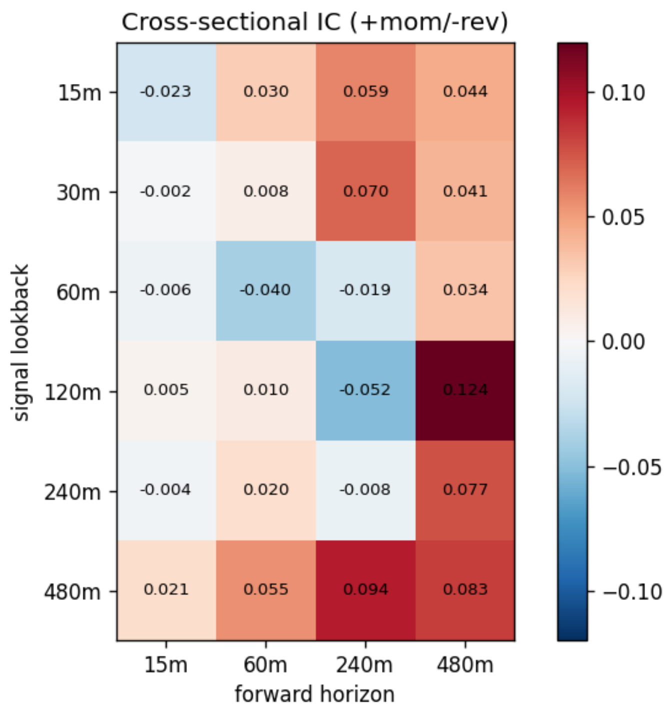
What I am proudest of is not a signal - it is that the system tested its own ideas on live data and overturned my own thesis. My backtest champion was momentum; the live feed showed momentum was dead and only weak mean reversion remained, so I retired it. I keep and publish the negative results rather than curve-fit around them. The edge is thin and regime-dependent - I say so plainly, and size for it. This pack is the evidence.

Inside: (1) the live data that overturned the thesis; (2) the overfitting controls I hold myself to; (3) the governed AI desk and its hard safety bounds.

1 - I let the data overturn my own thesis

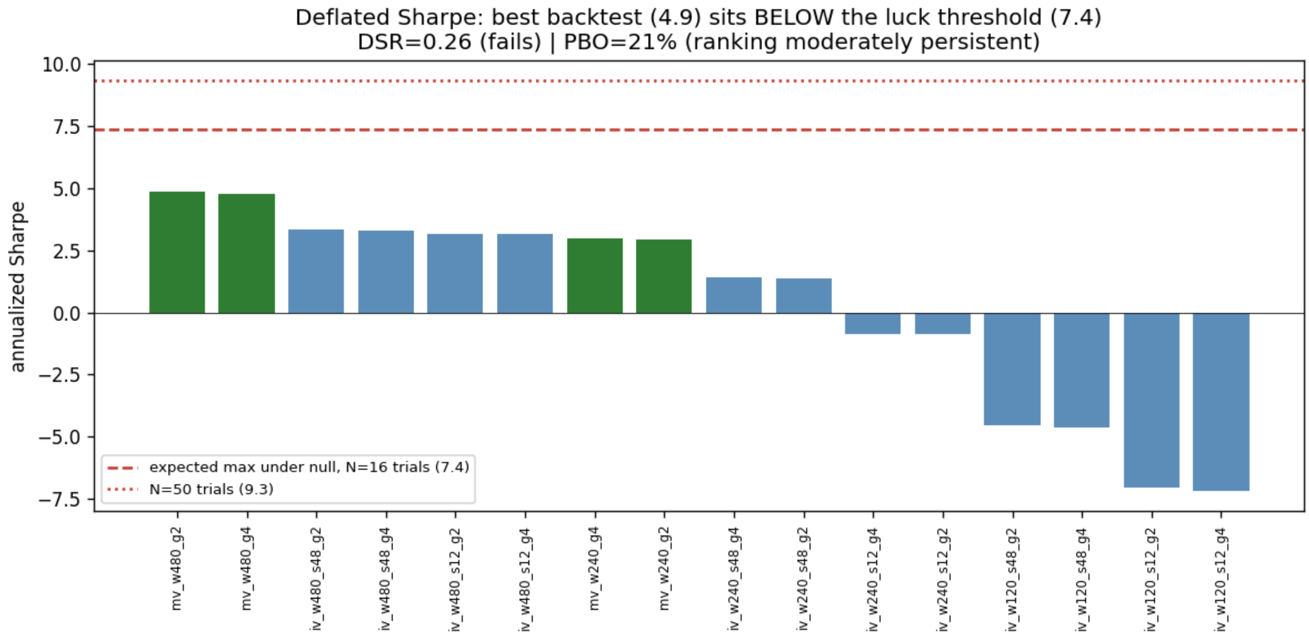


Live feed, 26 Jun (10 instruments, 6,606 one-minute bars). Returns are negatively autocorrelated at every horizon (-0.04 to -0.13, strongest at 60-120 min), while the momentum information coefficient is indistinguishable from zero. Translation: there is no momentum edge to chase, only weak short-horizon reversion - the reason I retired momentum live and wired a conservative reversion strategy.

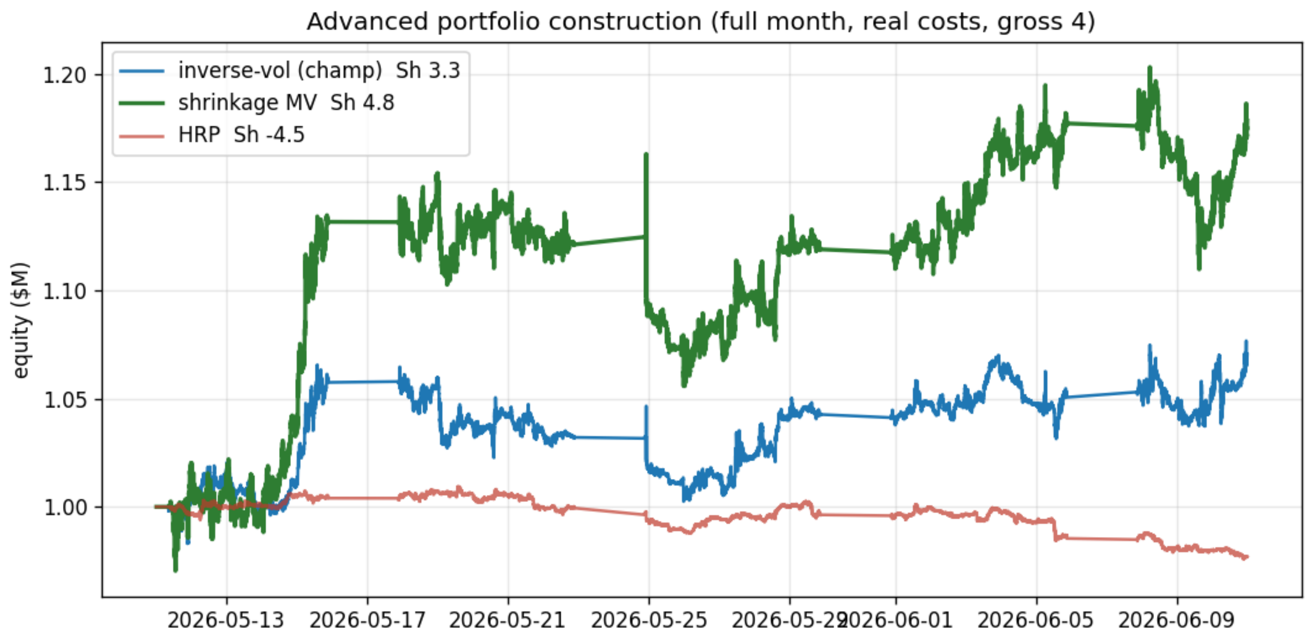


Cross-sectional IC surface (signal lookback x forward horizon). The edge is small and horizon-dependent - mostly |IC| < 0.06, with no stable momentum band. Consistent with a thin, regime-dependent signal, not a free lunch.

2 - Rigor: honesty as a feature

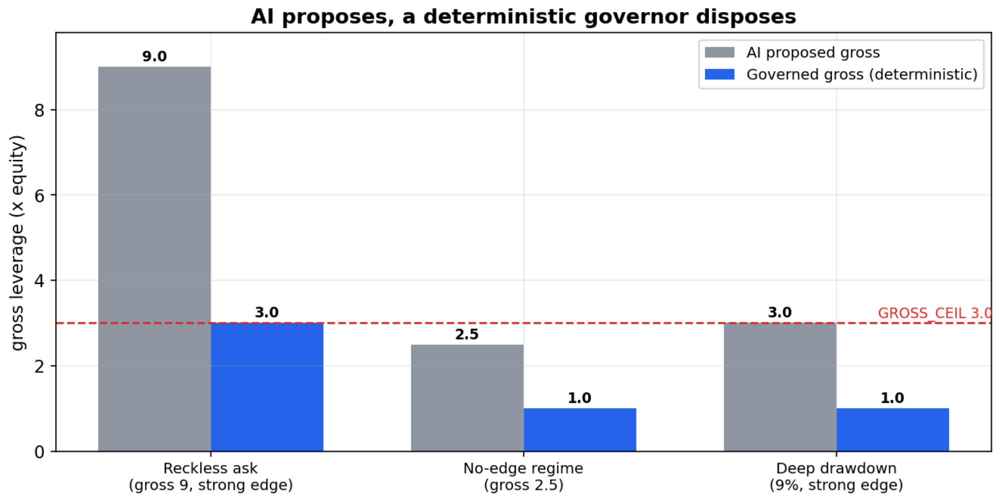


The honesty centerpiece. My best backtest Sharpe (4.9) sits below the 7.4 Sharpe expected from luck alone across 16 trials - Deflated Sharpe = 0.26 (fails), Probability of Backtest Overfitting about 21%. I report this rather than hide it: trust the direction of the findings, not the magnitude.



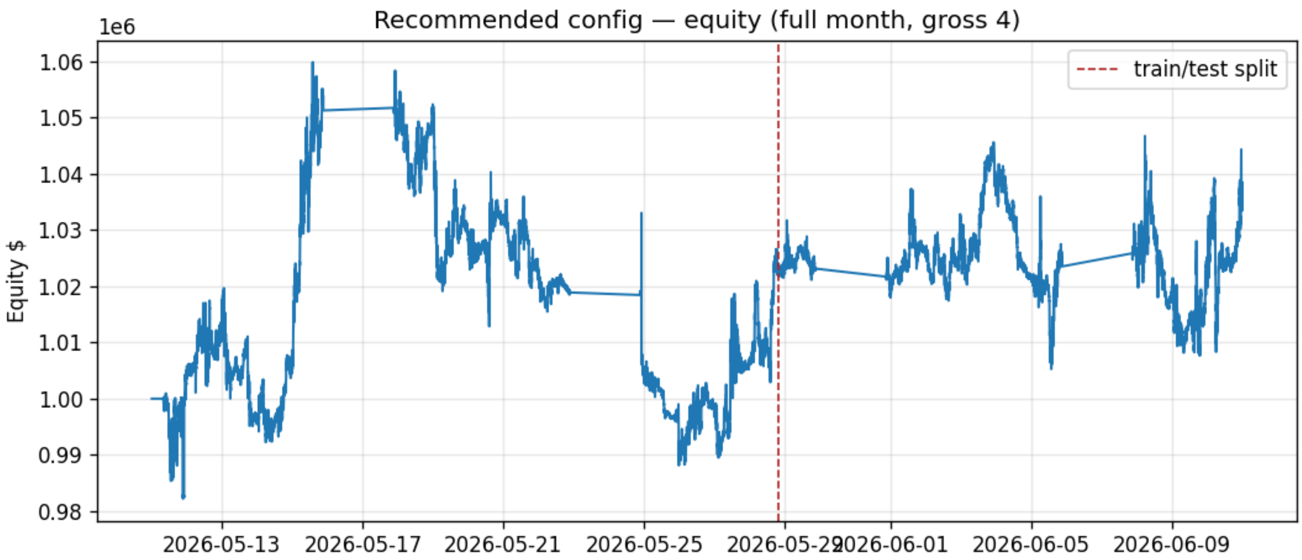
Portfolio construction under real transaction costs: shrinkage-covariance mean-variance (Sharpe 4.8) beat inverse-vol (3.3) and HRP (-4.5). It looks strong - but it should be read against the deflated-Sharpe panel above before believing the level.

3 - AI proposes, a deterministic governor disposes



Each case asserted in tests/test_safety.py (passing): reckless leverage capped at the 3.0 ceiling; a no-edge regime forced to 1.0; a 9% drawdown overrides even a strong edge to 1.0.

Every AI-proposed leverage is clamped to safe bounds before it can act. The three cases shown are exactly the assertions in tests/test_safety.py (passing): a reckless gross 9 is capped at the 3.0 ceiling, a no-edge regime is forced to 1.0, and a 9% drawdown overrides even a strong edge down to 1.0. The AI can advise, but it can never breach a risk limit.



Recommended config over the archive month with an out-of-sample split. Modestly positive and choppy, shown at face value. Paper/contest only; every AI overlay is reduce-only and cannot override the deterministic risk engine.